

MTH 310/520 Graph Theory: Quiz 2

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Total: 10 points

1) $G = (V, E)$ is called *elementary* iff $\forall e \in E \exists$ a perfect matching M for the graph G with $e \in M$ and G is connected.

Suppose G is a bipartite graph with bipartition (U, W) . Then prove the following are equivalent:

- a) G is elementary.
- b) G has exactly two minimum vertex covers, namely U and W .
- c) $|U| = |W|$ and for every non-empty proper subset X of U , $|N(X)| \geq |X| + 1$.
- d) $G = K_2$, or $|V(G)| \geq 4$ and for any $u \in U$, $w \in W$, $G \setminus \{u, w\}$ has a perfect matching.

[10 points]

[CO3,CO4]